MINIMAL SUFFICIENCY

THERE CAN BE MORE SUFFICIENT STATISTICS, EACH INDUCING A PARTITION OF Ω (SAMPLE SPACE). TWO SPECIAL CASES:

1). $T(\underline{X}) = \underline{X}$ (CONSIDER AS STATISTIC THE WHOLE DATASET) (NO DATA REDUCTION) => FINEST PARTITION WHERE EACH REALIZED VALUE OF $\underline{X}$ IS IN ITS OWN PARTITION SET

2). IF $T(\underline{x})$ IS SUFFICIENT, A ONE-TO-ONE FUNCTION OF $T(\underline{X})$ INDUCES SAME PARTITION OF $T(\underline{x})$ (IF THE TWO STATISTICS ARE RELATED MATEMATHICALLY IN A ONE-TO-ONE WAY)

WE WOULD LIKE TO REDUCE AS MUCH AS POSSIBLE THE INFORMATION I HAVE AND STILL NOT LOOSING ANY RELEVANT INFORMATION FOR θ

REDUCE AS MUCH AS POSSIBLE THE INFORMATION = INDUCING THE CLOSEST POSSIBLE PARTITION I CAN DO

WE SAY A SUFFICIENT STAT. $Y^* = t^*(\underline{X})$ IS BETTER THAN ANOTHER SUFF. STAT. $Y = t(\underline{X})$ IF Y^* INDUCES A LESS FINE PARTITION OF Ω

THIS IMPROVEMENT TYPICALLY HAPPENS WHEN Y^* IS A FUNCTION (NOT ONE-TO-ONE) OF Y. IN THIS CASE YOU MAY HAVE THAT $t^*(\underline{x}_1) = t^*(\underline{x}_2)$, BUT $t(\underline{x}_1) \neq t(\underline{x}_2)$

EX: $X_i \sim Bern(p)$ IID $i = 1, \dots, m$

SUFF. STAT. $Y^* = t^*(\underline{X}) = \sum_{i=1}^{m} X_i$ AND $Y = (t_1(\underline{X}), t_2(\underline{X}))$ (TWO DIMENSIONAL VECTOR), WHERE

$t_1(\underline{X}) = \sum_{i=1}^{m} X_i$, $m < m$

$t_2(\underline{X}) = \sum_{i=m+1}^{m} X_i$

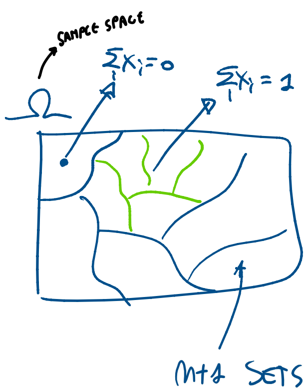
WE HAVE THAT

Y^* IS A FUNCTION OF Y

SINCE $t^*(\underline{x}) = t_1(\underline{x}) + t_2(\underline{x})$ $\Rightarrow$ Y^* ACHIEVES A HIGHER DATA REDUCTION

THE PARTITION INDUCED BY Y^* HAS $m+1$ SETS

WHILST " " BY Y HAS $(m+1) \cdot \underbrace{(m-m+1)}_{\text{POSSIBLE VALUES OF THE SUM}}$ SETS (FIRST COMPONENT OF Y)



⇒ THE SUFF. STATISTIC THAT INDUCES THE BEST PARTITION (MOST REDUCTION) IS THE MINIMAL SUFF. STATISTIC (FUNCTION OF ANY OTHER SUFF. STATISTIC)

A FUNCTION NOT ONE-TO-ONE OF ANY OTHER SUFFICIENT STATISTIC

WE CAN SAY THAT IN THE EXPONENTIAL FAMILY, THE SUFF. STATISTIC I SEE USING THE FACTORIZATION TH IS ALSO MINIMAL

GENERALLY SPEAKING: A STATISTIC IS SAID TO BE COMPLETE IF:

IF $\mathbb{E}(\overset{\text{some function}}{g}(T)) = 0 \ \forall\theta$ IMPLIES $\mathbb{P}(g(T)=0) = 1 \ \forall\theta$

⇒ $\left\{ \begin{array}{l} \text{COMPLETE} \\ + \text{SUFFICIENCY} \end{array} \right. \Rightarrow$ MINIMAL SUFFICIENCY

LAST POINT: MINIMAL SUFF. STATISTIC IS NOT UNIQUE

CAN HAVE MANY POSSIBLE STATISTICS WHICH ARE MINIMAL SUFFICIENT

(ANY ONE-TO-ONE FCN IS STILL MINIMAL SUFF)

POINT ESTIMATION METHODS

BASIC ASSUMPTION: DISTRIB. OF X CAN BE REPRESENTED BY PDF/PMF BELONG TO SOME PARAMETRIC STATISTICAL MODEL $\{ f(x;\theta), \theta \in \Theta \}$

LEARNING f REDUCES TO LEARNING θ

θ UNKNOWN ⇒ LOOKING AT THE DATA, I WANT TO ASSIGN TO θ A "REASONABLE" VALUE (POINT ESTIMATION)

⇒ WE USE AN ESTIMATOR OF θ: ANY SAMPLE STATISTIC I USE TO ESTIMATE θ (NOTE: ESTIMATOR IS RANDOM, SINCE FUNCTION OF $\underline{X}$)

VS ESTIMATE OF θ : THE REALIZED VALUE OF THE ESTIMATOR (NOTE: ESTIMATE IS NOT RANDOM, SINCE FUNCTION OF SMALL $\underline{x}$)

YOUR EX: ESTIMATE $\mathbb{P}(X>0)$

ESTIMATOR : $\frac{1}{m}\sum_{i=1}^{m} \mathbb{1}_{\{X_i>0\}}$ PROP. OF $X_i > 0$

ESTIMATE : $\underline{x} = (+1, -1, +1) \Rightarrow 2/3$ ESTIMATE

$$\mathbb{1}_{\{X_i>0\}} = \begin{cases} 1, & \text{IF } X_i > 0 \\ 0, & \text{OTHERWISE} \end{cases}$$ INDICATOR FUNCTION

WE WILL SEE
- METHODS TO FIND ESTIMATORS
- METHODS TO EVALUATE ESTIMATORS AND TO COMPARE THEM

METHODS TO FIND
- ANALOGY ESTIMATION
- METHOD OF MOMENTS (MM)
- MAXIMUM LIKELIHOOD ESTIMATION (MLE)
- BAYESIAN METHODS (LATER)

$\boxed{\text{ANALOGY METHOD}}$

INTUITION : ESTIMATE A POP QUANTITY WITH SAME QUANTITY BUT COMPUTED ON THE SAMPLE

EX : ESTIMATE POP MEAN WITH SAMPLE MEAN
" POP MEDIAN WITH SAMPLE MEDIAN
" POP PROPORTION WITH SAMPLE PROPORTION

EX: ANALOGY ESTIMATOR OF $\sigma^2 = \mathbb{E}[(X-\mu)^2]$ IS

POPULATION VARIANCE

$$\hat{\sigma}^2 = \frac{1}{m}\sum_{i=1}^{m}(X_i - \overline{X})^2$$

ESTIMATOR

SAMPLE MEAN

ANALOGY ESTIMATOR OF POPULATION VARIANCE

CARE: $\hat{\sigma}^2 \neq S^2$ AND SINCE $\mathbb{E}(S^2) = \sigma^2$, THEN

SAMPLE VARIANCE

$$\mathbb{E}(\hat{\sigma}^2) = \mathbb{E}\left(\frac{m-1}{m-1}\sum_{i=1}^{m}(X_i-\overline{X})^2 / m\right) = \mathbb{E}\left(\frac{m-1}{m}S^2\right) = \frac{m-1}{m}\mathbb{E}(S^2)$$

$$= \frac{m-1}{m}\sigma^2$$

THERE ARE NO GUARANTEES THAT ANALOGY ESTIMATION WILL WORK WELL

$\boxed{\text{METHOD OF MOMENTS}}$ (MM)

(SEMI-PARAMETRIC METHOD)

↳ YOU DO NOT NEED THE WHOLE PDF/PMF OF X, BUT JUST FEW MOMENTS

IDEA: WE EQUATE POPULATION MOMENTS AND SAMPLE MOMENTS

↳ $\mathbb{E}(X), \mathbb{E}(X^2), \mathbb{E}(X^3), \dots$

$\frac{1}{m}\sum_{i=1}^{m} X_i, \quad \frac{1}{m}\sum_{i=1}^{m} X_i^2, \quad \dots$

FIX

$$\begin{cases} \mathbb{E}(X) = \overline{X} \\ \mathbb{E}(X^2) = \frac{1}{m}\sum_i X_i^2 \\ \vdots \\ \mathbb{E}(X^p) = \frac{1}{m}\sum_i X_i^p \end{cases}$$

HOW MANY EQUATIONS? IT DEPENDS ON THE DIM OF θ!

EX:

- $X \sim Exp(\lambda) \Rightarrow 1$ EQUATION: $\mathbb{E}(X) = \overline{X}$ ENOUGH
- $X \sim N(\mu, \sigma^2) \Rightarrow 2$ EQS $\begin{cases} \mathbb{E}(X) = \overline{X} \\ \mathbb{E}(X^2) = \overline{X}_2 \end{cases}$

  TWO PARAMETER TO ESTIMATE

NOTE: LHS OF EQS DEPEND ON θ

LEFT HAND SIDE

↳ SOLVE THE SYSTEM FOR $\theta \Rightarrow$ YOUR MM ESTIMATOR OF θ

METHOD OF MOMENT

NOTE 2: EQS MAKE SENSE BECAUSE OF LLN

LAW OF LARGE NUMBERS

EX: $X \sim N(\mu, \sigma^2)$, BOTH μ, σ^2 UNKNOWN.

$\theta = (\mu, \sigma^2) \Rightarrow$ IN MM (METHOD OF MOMENT) 2 EQS

$$\begin{cases} \mathbb{E}(X) = \bar{X} \\ \underbrace{\mathbb{E}(X^2)}_{V(X)+\mathbb{E}(X)^2} = \boxed{\frac{1}{m}\sum_i X_i^2} \; \bar{X}_2 \end{cases} ; \quad \begin{cases} \tilde{\mu} = \bar{X} \\ \tilde{\sigma}^2 + \tilde{\mu}^2 = \frac{1}{m}\sum_{i=1}^m X_i^2 \end{cases}$$

FROM $V(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$

$\Rightarrow$ MM ESTIMATOR OF μ IS $\bar{X}$

$\Rightarrow$ MM ESTIMATOR OF σ^2 IS

$$\tilde{\sigma}^2 = \frac{1}{m}\sum_{i=1}^m X_i^2 - \tilde{\mu}^2 = \frac{1}{m}\sum_i X_i^2 - \bar{X}^2$$

$$= \frac{1}{m}\sum_i \left[\left((X_i - \bar{X}) + (\bar{X}) \right)^2 \right] - \bar{X}^2$$

$$= \frac{1}{m}\sum_i \left[(X_i - \bar{X})^2 + \bar{X}^2 + 2\bar{X}(X_i - \bar{X}) \right] - \bar{X}^2$$

$$= \frac{1}{m}\sum_i (X_i - \bar{X})^2 + \cancel{\underbrace{\frac{1}{m}\sum_i \bar{X}^2}_{\bar{X}^2}} + \cancel{2\bar{X}\frac{1}{m}\underbrace{\sum_i (X_i - \bar{X})}_{=0}} - \cancel{\bar{X}^2}$$

$$\sum_i X_i - \sum_i \bar{X}$$

$$\frac{m}{m}\sum_i X_i - m\bar{X}$$

$$\bar{X}$$

MM (IN THIS EX) IS EQUAL TO ANALOGY

$$= \frac{1}{m}\sum_i (X_i - \bar{X})^2$$

$$= \frac{m-1}{m} S^2$$

EX (MM FAILS)

$X \sim Bin(N, p)$, BOTH N AND p UNKNOWN

$\underline{X} = (X_1, \dots, X_m)$ RANDOM SAMPLE $\Rightarrow$ IN MM WE NEED 2 EQS

$$\begin{cases} \mathbb{E}(X) = \overline{X} \\ \underbrace{\mathbb{E}(X^2)}_{V(X)+\mathbb{E}(X)^2} = \overline{X}_2 \end{cases} \; ; \quad \begin{cases} \tilde{N}\tilde{p} = \overline{X} \\ \tilde{N}\tilde{p}(1-\tilde{p}) + (\tilde{N}\tilde{p})^2 = \overline{X}_2 \end{cases}$$

... SOLVE FOR $\tilde{N}$ AND $\tilde{p}$ ...

$$\begin{cases} \tilde{N} = \dfrac{\overline{X}^2}{\overline{X} - \frac{1}{m}\sum_i (X_i - \overline{X})^2} \\ \tilde{p} = \overline{X}/\tilde{N} \end{cases}$$

VALUES RESONABLE FOR BINOMIAL

$\underline{X} = (1, 5, 4, 1)$ $\quad m = 4$

$$\Rightarrow \begin{cases} \tilde{N} = -3.469 \\ \tilde{p} = -0.9286 \end{cases}$$

THE NUMBER OF BERNULLI TRAIALS CAN NOT BE NEGATIVE :(

$\Downarrow$

MM FAILS

$\Uparrow$

A PROBABILITY CAN NOT BE NEGATIVE :(